3DBionotes COVID-19 Edition

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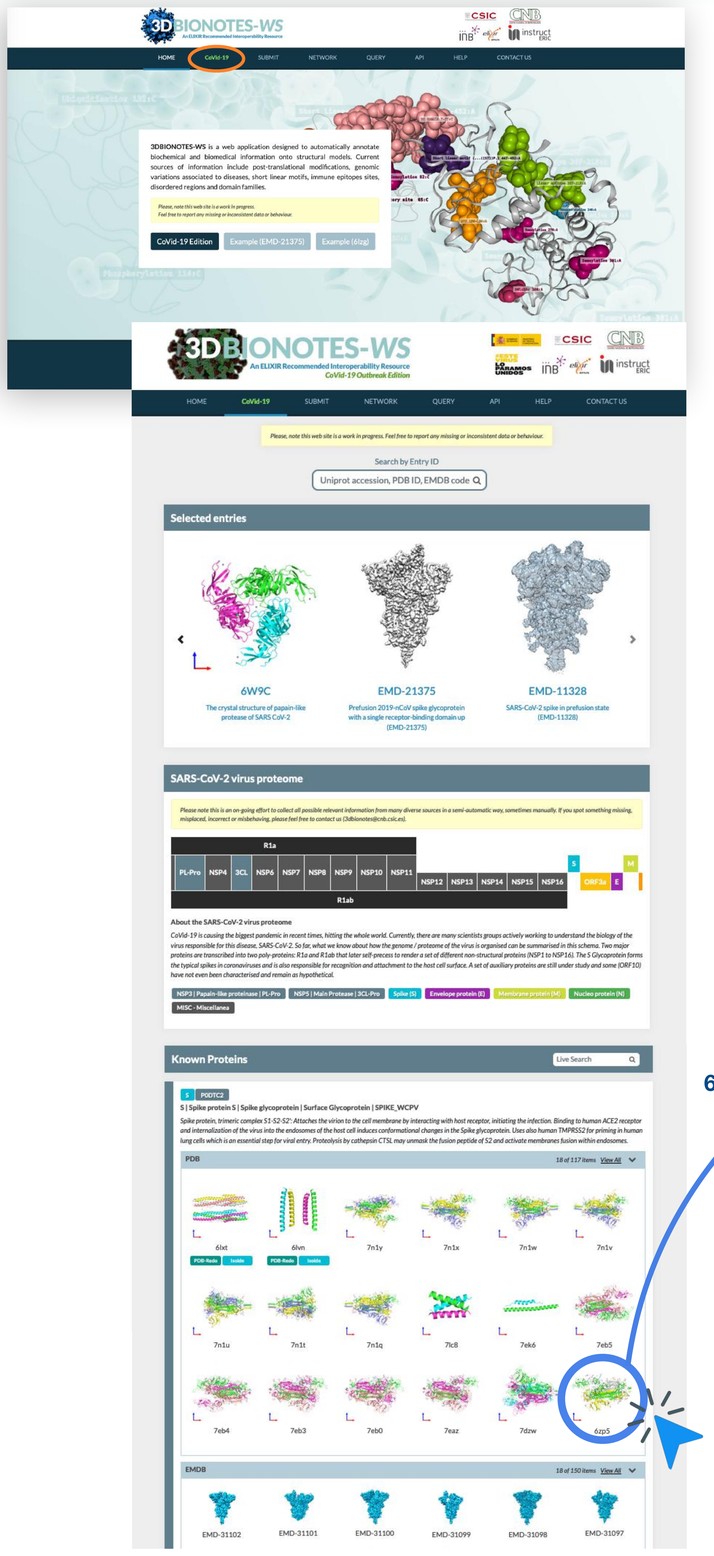
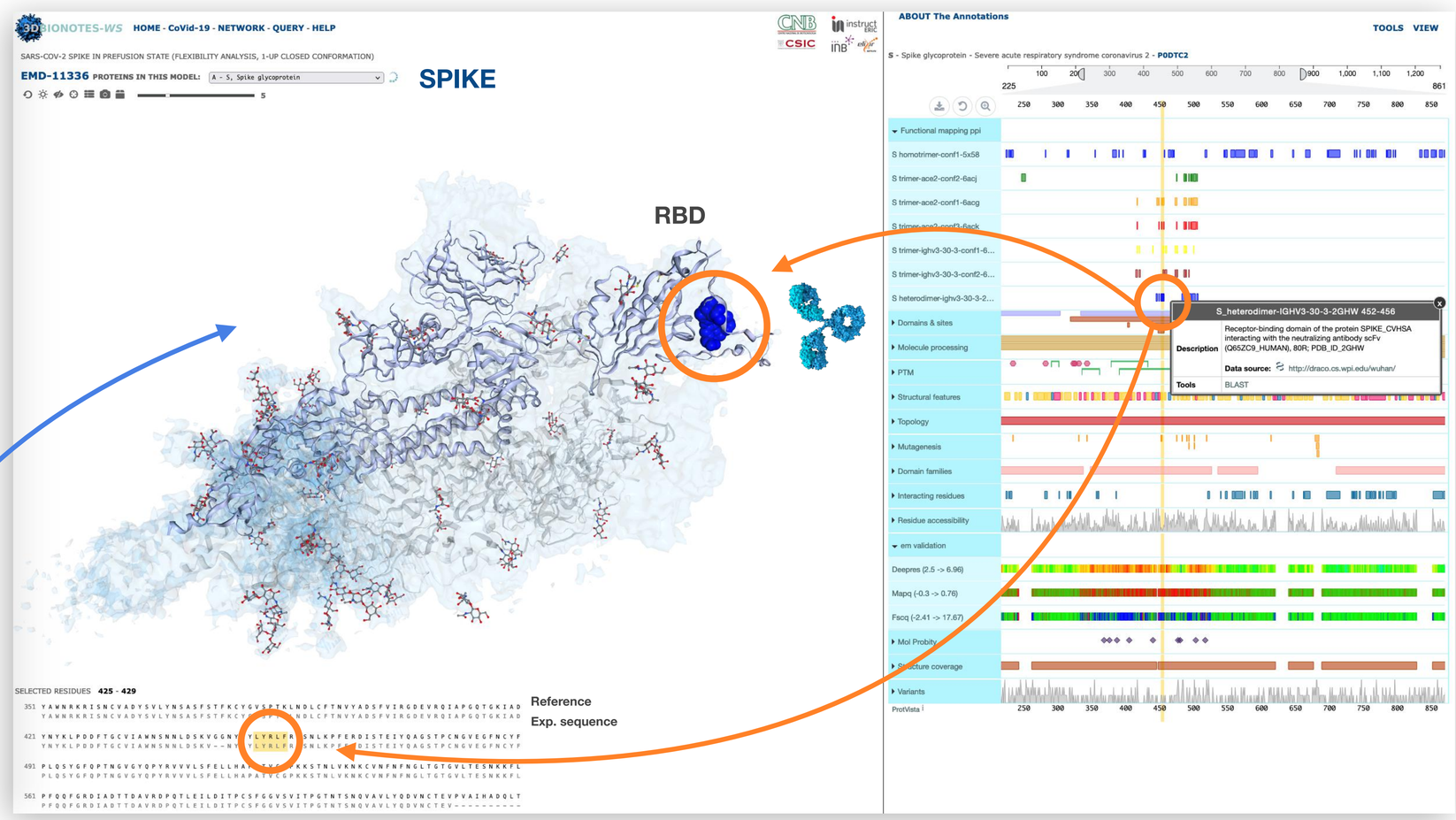
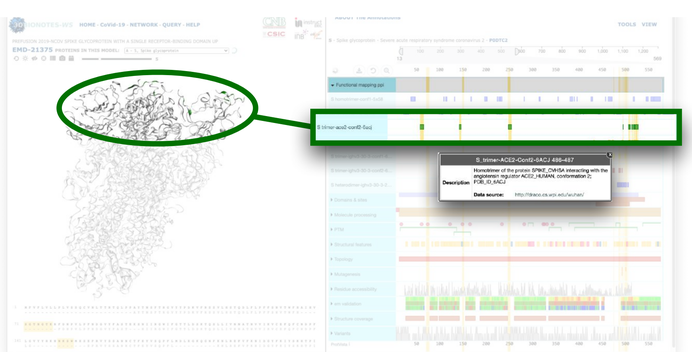


3DBIONOTES-WS

3DBionotes-WS is a set of Web Services providing multiple annotations oriented to structural biology analysis, plus a website interface including a 3D viewer for macromolecular models and Cryo-EM derived maps. The display is fully interactive with a corresponding display for genomic, proteomic and structural feature annotations. With the COVID-19 outbreak, easy access to quality data for the required structural analysis of SARS-CoV-2 and other related viruses and its interactions with human macromolecules become a necessity. We have set up a new section in the website aiming to collect and provide all available structural information available.

There, we have a panel with a brief description and its role in the virus life cycle. This panels contain all known structures organized in 5 sections: the atomic models published in the Protein Data Bank (PDB), volumen maps from the Electron Microscopy Database (EMDB), interactions with other proteins of the virus or with the host proteins, with small molecules (ligands) that have been identified in drug screening studies, and for completeness, related proteins from other coronavirus (SARS1, MERS, etc) are also collected.

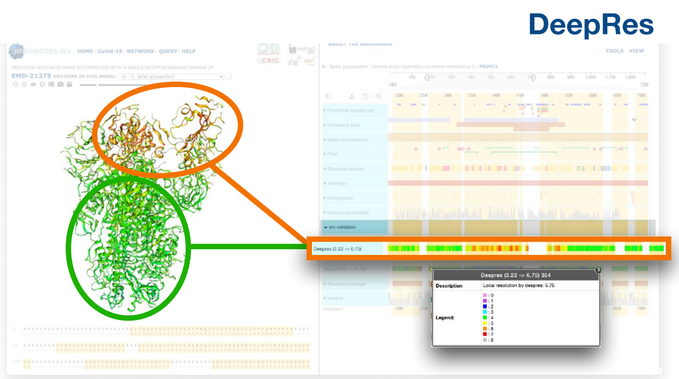


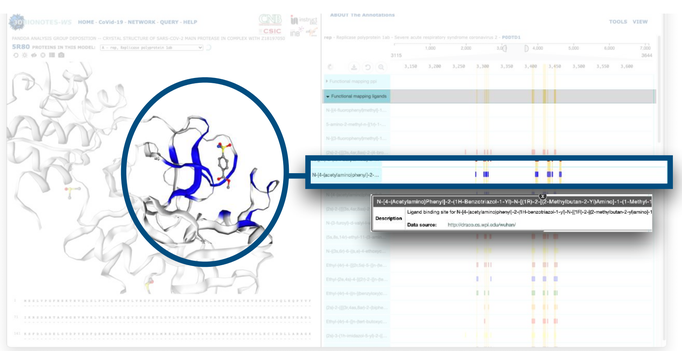


FUNCTIONAL MAPPINGS

Functional mapping tracks allow to locate the residues that likely constitute **protein-protein interaction (PPI) binding sites** (GREEN) and **ligand binding sites** (BLUE). These mapped residues have been obtained by homology modeling. All those interacting amino-acids are putative functional sites and have to be validated experimentally.

Drug screening track displays the different **binding sites** (RED) **experimentally** obtained at Diamond synchrotron (UK) in the XChem fragment screening campaign against SAR-CoV-2 Main Protease and Nsp3-macrodomain.

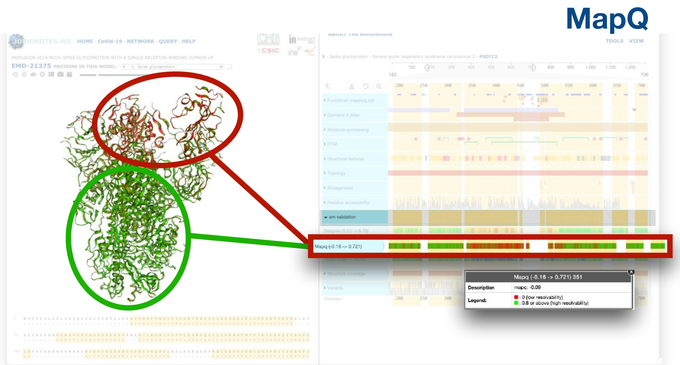


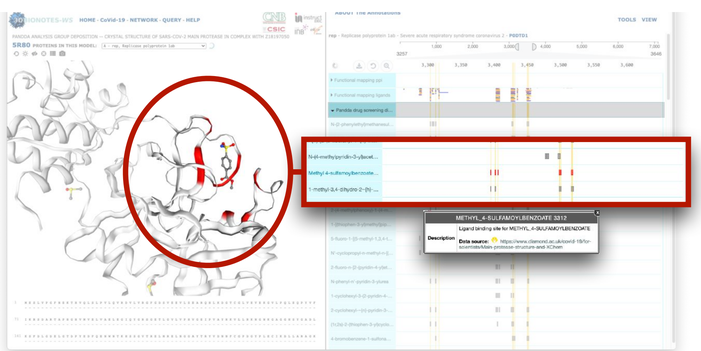


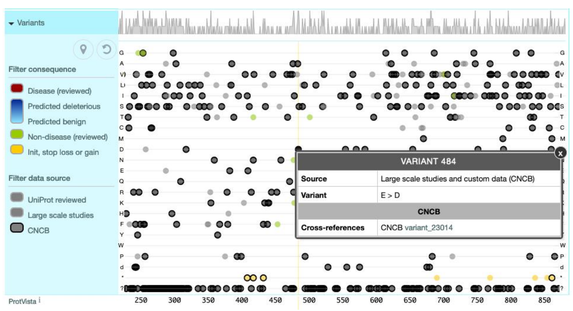
CRYO-EM VALIDATIONS

Local resolution for cryo-EM maps is calculated applying different algorithms (**DeepRes** and MonoRes). If there is an atomic model available in PDB, **MapQ** and **FSC-Q** scores can be calculated. The local resolution values are projected onto the atoms of the atomic model, using convenient color codes according to the different thresholds. For instance, in FSC-Q red indicates atoms that may be associated with noise (possible overfitting) and blue indicates poorly fitted atoms or areas with low resolvability.

This way is quite easy to find interesting regions with similar resolution values

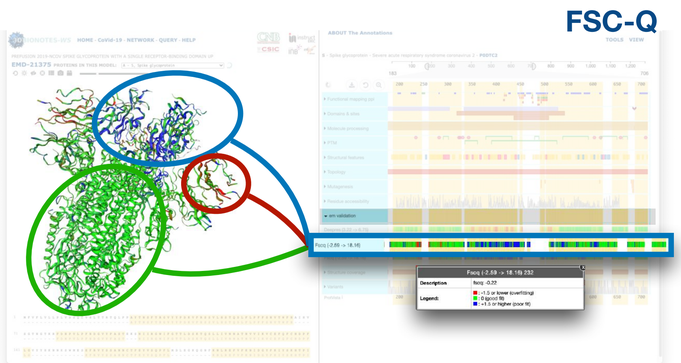






VARIANTS

Variants track summarizes the aggregated observed mutations at a certain position in the protein that have been compiled at the China National Center for Bioinformatics and Uniprot.



GOBIERNO DE ESPAÑA · MINISTERIO DE CIENCIA E INNOVACIÓN · CSIC Consejo Superior de Investigaciones Científicas · CNB Centro Nacional de Biotecnología · BIO Computing Unit · instruct Image Processing Center · instruct ERIC · inb · elixir SPAIN